



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 10

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

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Semester / Year:

Academic Session:

Date of Performance:

Date of Submission:

CODE:

M ~

```
ishak@YASHAWINI MSYS ~  
$ nano SSTF.c
```

M ~

```
GNU nano 8.7 SSTF.c  
#include <stdio.h>  
#include <stdlib.h>  
  
int main() {  
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};  
    int n = 8;  
    int head = 53;  
    int total_movement = 0;  
  
    int visited[8] = {0}; // to track serviced requests  
  
    for(int i = 0; i < n; i++) {  
        int min_distance = 1000;  
        int index = -1;  
  
        // Find closest request  
        for(int j = 0; j < n; j++) {  
            if(!visited[j]) {  
                int distance = abs(head - requests[j]);  
                if(distance < min_distance) {  
                    min_distance = distance;  
                    index = j;  
                }  
            }  
        }  
  
        // Move head  
        total_movement += min_distance;  
        head = requests[index];  
        visited[index] = 1;  
    }  
  
    printf("Total Head Movement = %d\n", total_movement);  
  
    return 0;  
}
```

OUTPUT:

```
ishak@YASHAWINI MSYS ~  
$ gcc SSTF.c -o SSTF && ./SSTF  
Total Head Movement = 323  
  
ishak@YASHAWINI MSYS ~  
$
```